ASSIGNMENT I

MUSEUMS AQUARIUMS AND ZOOS

DATA SCIENCE WITH R PROJECT

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Dataset Introduction

01

About the Dataset

To begin this report, we will start by introducing some basic facts about the chosen dataset: the Museums, Aquariums, and Zoos Dataset including its source, context, size, variables, variable formats and so on. We will later refer to it as the Museums Dataset for simplicity since aquariums and zoos are identified as a type of museums in the dataset.

- Source:

The museums dataset was downloaded from Kaggle.com, I started by searching for the keywords I was interested in on different dataset sites, such as "museums", "art", "galleries", etc. and eventually found this dataset on Kaggle.

- Context:

This dataset is a list of museums, aquariums, zoos and related organizations in the United States. It includes basic information about each organization (name, state, city, income...) as well as the type or discipline of the organization.

- Size:

After the data cleaning process, we ended up with 10,979 observations and 8 variables, which contained data for 9 types of museums, spread over 7,236 cities in 51 states of the United States.

Below are the names of each variable, the formats of the variable and the number of categories in each variable that we have retained for further analysis:

Variable Name	Variable Format	Categories Number
MuseumID	num	10979
MuseumName	chr	10979
MuseumType	factor	9
Address	chr	10979
City	factor	3942
State	factor	51
Income	num	10979
Revenue	num	10979

Dataset Introduction

01

Data Cleaning

Before starting the analysis, it is necessary to check the accuracy and completeness of the data and to remove unneeded data. The detailed data cleaning process is outlined below:

Step 1: Retain the desired variables and remove the unrelated ones

E.g., there are "Museum Name", "Legal Name", and "Alternate Name" which all indicate the name of the museum with minimal differences so here we only retained the "Museum Name". By the end of this step, we went from 25 to only 8 variables.

Step 2: Change column names to short, relevant names with no spaces

E.g., "City (Administrative Location)" to just "City".

Step 3: Check if there is repeated observations and remove the duplicates

There ware repeated museum names and after delete the duplicates we got from 33,072 to 31,546 rows.

Step 4: Check if there is missing value and remove the observations with missing values

Through summary we could see that only "Income" and "Revenue" columns have N/A values, as we cannot accurately replace the missing data, we will discard the relevant rows. At this point we went from 31,546 to 21,155 rows.

Step 5: Remove all the observations with Income or Revenue equals to 0

Scrolling down the data at this time, we saw that there are a lot of museum with value 0 for their income or revenue, as these 2 are essential data to be analyzed afterwards, we assumed that these 0 values represent incorrect observations and deleted these rows. After doing that we went from 21,155 to 10,979 rows.

Step 6: Check if all variables are in the right format

Till now we have 2 formats, 3 num variables and 4 chr variables.

Step 7: Add 3 categorical variables - MuseumType, City and State

Change the format of these 3 variables from chr to factor in order to facilitate the analysis. By checking the numbers of categories, we can summarize that we have data for 9 types of museums, spread over 7,236 cities in 51 states.

You can find below the final cleaning result of the database:

MuseumID	MuseumName	MuseumType	Address	City	State	Income	Revenue
8400200098	ALASKA AVIATION HERITAGE MUSEUM	HISTORY MUSEUM	4721 AIRCRAFT DR	ANCHORAGE	AK	602912	550236
8400200117	ALASKA BOTANICAL GARDEN	ARBORETUM, BOTANICAL GARDEN, OR NATURE CENTER	4601 CAMPBELL AIRSTRIP RD	ANCHORAGE	AK	1379576	1323742
8400200153	ALASKA CHALLENGER CENTER FOR SPACE SCIENCE TE...	SCIENCE & TECHNOLOGY MUSEUM OR PLANETARIUM	9711 KENAI SPUR HWY	KENAI	AK	740030	729080
8400200027	ALASKA HERITAGE MUSEUM	HISTORY MUSEUM	301 W NORTHERN LIGHTS BLVD	ANCHORAGE	AK	602912	550236
8400200078	ALASKA JEWISH MUSEUM	GENERAL MUSEUM	1117 E 35TH AVE	ANCHORAGE	AK	2658938	34374
8400200084	ALASKA LIGHTHOUSE ASSOCIATION	HISTORIC PRESERVATION	2116 B 2ND ST	DOUGLAS	AK	16500	16500
8400200073	ALASKA MINING HALL OF FAME FOUNDATION	HISTORY MUSEUM	PO BOX 81906	FAIRBANKS	AK	184295	31393
8400200123	ALASKA NATIVE HERITAGE CENTER	GENERAL MUSEUM	8800 HERITAGE CENTER DRIVE	ANCHORAGE	AK	4502727	4050394
8400200082	ALASKA SPORTS HALL OF FAME	GENERAL MUSEUM	6740 SAMUEL CT	ANCHORAGE	AK	1370930	615273

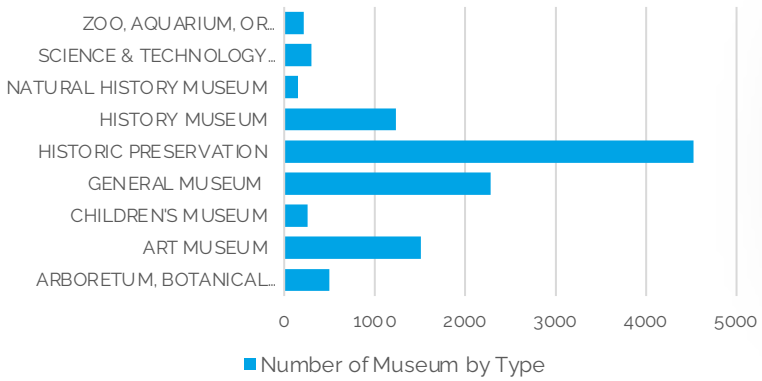
Dataset Identity

02

Simple Statistics

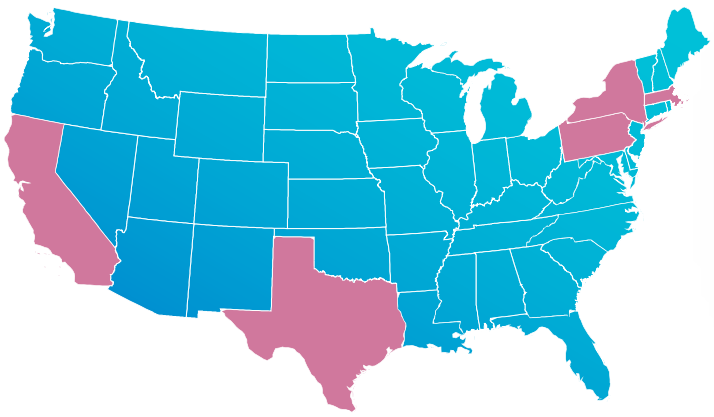
With a basic analysis on the dataset in R, we obtained some simple statistics as follow:

Number of Museum by Type



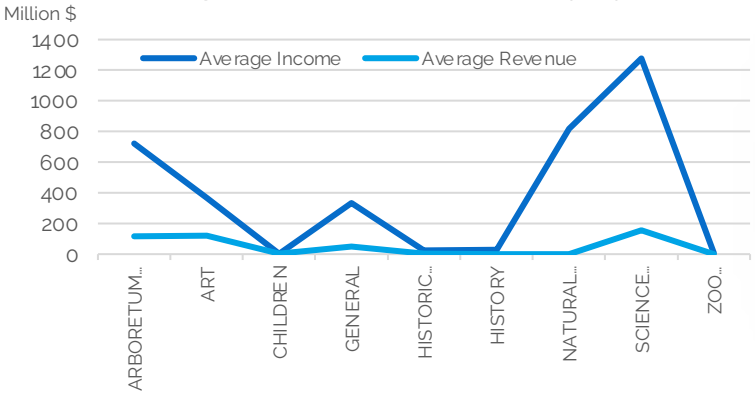
Historic Preservation is the most numerous museum types in the United States (41.2%), while **Natural History Museums** are the least represented of all types (1.4%).

Top 5 States with the most museums



The top five states with the highest number of museums in the United States in order are **California, New York, Pennsylvania, Texas, and Massachusetts.**

Avg Income and Revenue by Type



The museum type with the highest average income is **Science & Technology Museum or Planetarium** and the lowest is **Children's Museum**. Same situation with average revenue.

Storytelling

03

Story & Purposes

Having grown up with parents who are both art-related practitioners, painting has always been my biggest passion and I have always enjoyed visiting museums, especially art museums. They are great places to appreciate beauty and creativity where gives me a glimpse into different cultures, art and history. At the same time, I am also an avid lover of wildlife. Visiting local museums, aquariums and zoos is all the time a must-do for me when traveling to new destinations. I was therefore very excited when I found this dataset that was perfectly suited to my interests.

The general idea behind this database is to extract insights about museums and related organizations in the United States. Although the composition of this dataset is simple, we can still draw considerable inspirations from it for possible decision making. For example, how the composition of museum types varies across the United States, whether there are disparities in the ability of different types of museums to generate profits, whether there are significant gaps in the number of museums and income levels between each state or indeed city, and as to whether there are distinct country-specific factors behind these differences.

In a nutshell, we will look at the number, revenues and incomes of museums in the United States, and analyzing them along three dimensions: museum type, city and state.

- Extra goal:

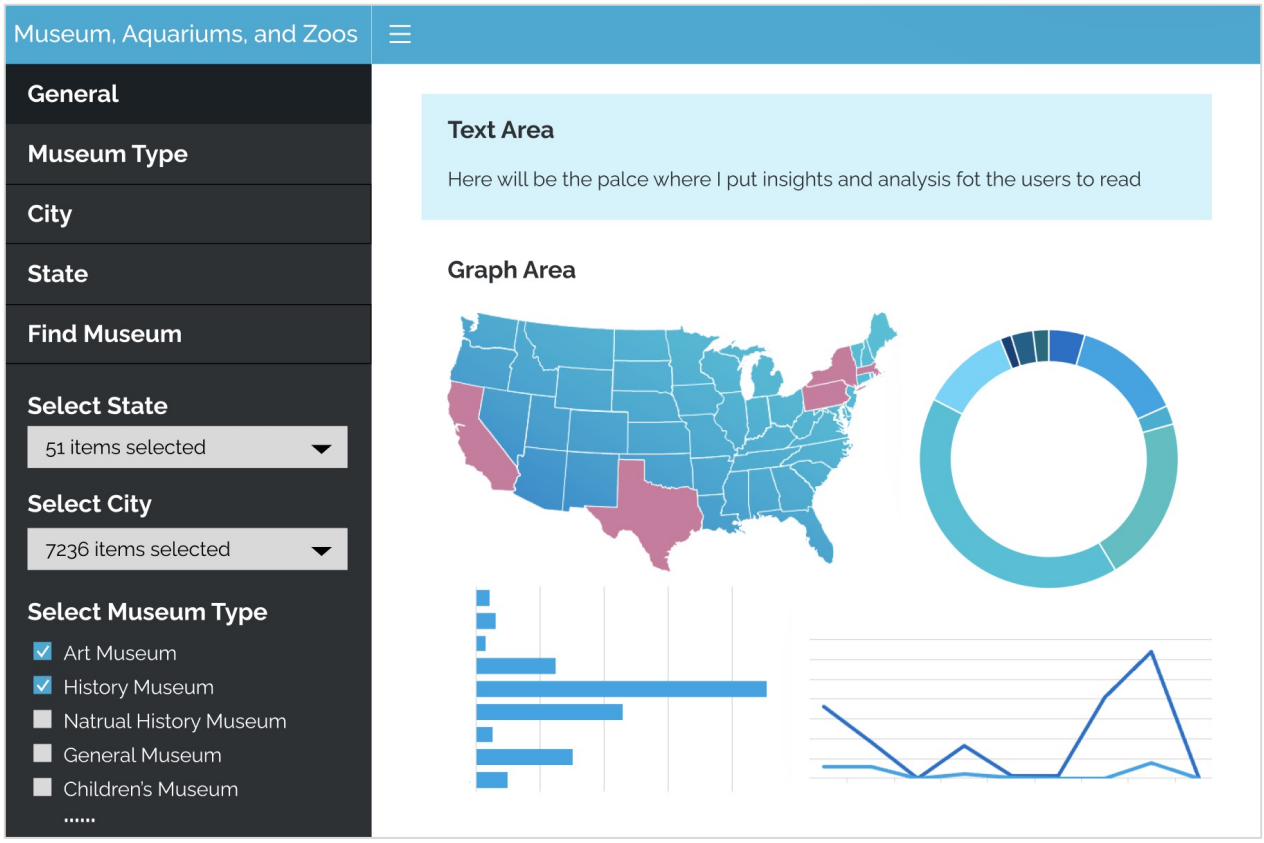
In addition to generating insights, I would also like to use this dataset for a function that filters museums based on location and type of museum. The usage scenario could be, when users are traveling to the United States and wish to visit local museums, they can select the destination state or city they are going, and filter the types of museums they are interested in (art, history, nature...). Then the users will be able to access a list of the basic information (name, type, address, city and state) of the museums filtered based on their options. Which is why I keep the "Address" variable in the clean dataset as well.

Although we removed roughly two-thirds of the museum data in the data cleaning process so this does not cover 100% of the museums in the United States, this dataset does contain a very large number of small private galleries, studios, etc. that might not be found on TripAdvisor, so perhaps it could bring some surprising results! (At least I'd like to try it myself!)

Application Draft

04 Imagined Application

Below is the outlook sketch of the imagined application I made by Figma:



On the left will be the menu bar where users can access to different dimensions of the analysis, that is, museums number, revenues and incomes analysis by museum type, city, state or in general. There will also be a "Find Museum" section where users can filter the museums they are interested in. Underneath this section will be the 3 options for users to choose and modify, with an increasingly precise logic. As for states and cities, there are a huge number of items to choose from and I will take this into account and try to find a simpler solution for the final version, for example, perhaps having the user select or enter the initial letter of the target item first and then swipe to choose.

The right hand side will therefore be the results display area. The top section will be where I will put the insights and analysis in text format for the user to read. And the bottom will be where I display different graphs of the statistics. Ideally, these graphics will also change in accordance with the options selected by the user on the left.